

Environment Setup — Workshop UPNA 2026

FHIR, REST & Web Applications

This document explains how to prepare your environment for the workshop.

Please read this carefully and complete the **pre-check before Day 1**.

What You Really Need (Quick Summary)

For Day 1 (FHIR + REST labs):

You need ONE of the following:

- Google Colab (recommended)
- Local Python + Jupyter Lab
- VS Code + Jupyter extension
- Jupyter.org (browser-based execution)

For Day 2 (Web App demo):

You need ONE of the following:

- Python installed (to run `python -m http.server`)
- VS Code with “Live Server” extension

If you work locally (Jupyter Lab or VS Code), you should also have:

- Git installed (recommended)

Optional REST clients:

- Postman
- VS Code + REST Client extension
- <https://hoppscotch.io/>

If you can:

- Run a notebook
- Execute Python code
- Reach the FHIR server
- Start a small local web server
- Access the workshop repository

→ You are fully prepared.

Getting the Workshop Materials

The workshop materials are provided via a Git repository.

If you are using Google Colab, a direct link to open the notebooks will be provided.

If you are working locally, you must download the repository.

Option A — Using Git (Recommended)

Check if Git is installed:

```
git --version
```

If Git is not installed:

Download from: <https://git-scm.com/>

To clone the repository:

```
git clone https://github.com/rafamayo/Workshop-UPNA-2026.git
```

Then navigate into the folder:

```
cd <repository-folder>
```

Option B — Download ZIP (Fallback)

If you prefer not to use Git:

1. Open the repository in your browser.
2. Click "Code" → "Download ZIP".
3. Extract the ZIP file.
4. Open the extracted folder in Jupyter or VS Code.

Note:

- If materials are updated, you will need to re-download the ZIP.
 - Git makes updates easier.
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Day 1 — FHIR & REST (Notebook-Based)

You must be able to run Jupyter notebooks.

Choose ONE of the following setups.

Option 1 — Google Colab (Recommended)

No installation required.

Steps:

1. Visit: <https://colab.research.google.com>
2. Log in with a Google account.
3. Create a new notebook.

4. Run:

```
print("Colab is working!")
```

If it runs without error → you are ready.

Why recommended:

- No installation
 - No dependency problems
 - Works in any browser
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Option 2 — Local Installation (Jupyter Lab)

Requirements:

- Python 3.10 or newer
- pip installed
- You are in the repository folder
- Create a virtual environment

```
python -m venv ./myvenv
```

1. Activate the virtual environment

```
source ./myvenv/bin/activate
```

1. Install the required modules:

```
pip install jupyter requests pandas matplotlib
```

1. Start Jupyter:

```
jupyter lab
```

1. Open the workshop notebooks and run:

```
print("Local Jupyter works!")
```

Caution: Every time you want to work on the notebooks you need to

1. Activate the virtual environment

```
source ./myvenv/bin/activate
```

1. Start Jupyter:

```
jupyter lab
```

Using a virtual environment makes sure that your project doesn't mess with the rest of the system!

Option 3 — VS Code + Jupyter Extension

Requirements:

- Python installed
- VS Code installed

Install extension:

- "Jupyter" (by Microsoft)

Open the notebook file in VS Code and run:

```
print("VS Code Jupyter works!")
```

Option 4 — Jupyter.org (Fully Browser-Based)

Visit:

<https://jupyter.org/try>

Upload the notebook file.

Run:

```
print("Remote Jupyter works!")
```

Note:

- This environment may reset
- Files may not persist
- Web app demo will still require local server on Day 2

Use only if other options are not possible.

Day 1 — FHIR Server Pre-Check

Before the workshop, test connection to the FHIR server.

In your notebook environment, run:

```
import requests
```

```
FHIR_SERVER = "https://hapi.fhir.org/baseR4/"  
r = requests.get(FHIR_SERVER + "metadata")  
print(r.status_code)
```

Expected result:

- 200 → OK
- JSON output → Perfect

If you see errors:

- Check internet connection
 - Check firewall restrictions
 - Try another network
 - Inform the instructor
-

Day 2 — Web App Demo

For Day 2, you must be able to run a small local web server.

Choose ONE option.

Option 1 — Python Lightweight Server (Recommended)

Navigate to:

labs/day2/webapp-demo/webapp-skeleton/

Start server:

macOS / Linux:

```
python3 -m http.server 8000
```

Windows:

```
py -m http.server 8000
```

Open in browser:

<http://localhost:8000/>

If you see the web page → ready.

Option 2 — VS Code + Live Server

Requirements:

- VS Code installed
- “Live Server” extension installed

Steps:

1. Open index.html

2. Right-click
3. Select "Open with Live Server"

Browser should open automatically.

Optional: REST Clients

During the workshop you may want to test REST calls outside Python.

You can use:

- Postman (desktop app)
- VS Code + REST Client extension
- Hoppscotch (online): <https://hoppscotch.io/>

These are optional and not required.

Troubleshooting

Cannot reach FHIR server

- Check URL carefully
 - Ensure internet access
 - Try different browser
 - Try different network
 - Instructor will provide fallback endpoint if needed
-

Git not found

Check:

```
git --version
```

If not installed:

<https://git-scm.com/>

Python not found

Check:

Windows:

```
py --version
```

macOS/Linux:

```
python3 --version
```

Install from:

<https://www.python.org/>

Final Check Before Workshop

You are ready if:

- You can run a notebook cell
- You can send a GET request to the FHIR server
- You can start a local web server
- You can open <http://localhost:8000/>
- You have access to the workshop repository

Test everything at least one day before the workshop.